Hyuksoon Chang

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Education

M.S. **Pohang University of Science and Technology** — Aug 2023 – Present
Computer Science and Engineering (advised by Kyungmin Bae)

B.S. **Chungbuk National University** — Mar 2019 – Mar 2023
Computer Engineering (advised by Kyuri Jo))

Publications

Model Checking of WebAssembly Software Using Maude — 2025
Hyuksoon Chang , Kyungmin Bae. Korea Computer Congress(2025)

Robustness Analysis of Convolutional Neural Networks for Circle Image Classification by Explainable Artificial Intelligence — 2021
Jungho Park*, **Hyuksoon Chang*** , Kyuri Jo. Korea Computer Congress(2021)

Inferring transcriptomic cell states and transitions from time series transcriptome data — 2021
Jo K, Sung IY, Lee D, **Jang H**, Kim S. Scientific Reports(2021)

Projects & Experience

Model Checking of WebAssembly Software with WASM-DSL — 2024 - Present

- Developing a model-checking framework for WebAssembly software to verify distributed-system properties
- Implemented the official WebAssembly specification in the Maude rewriting-logic framework to execute Wasm modules and perform LTL-based model checking.
- Designed compilation rules that translate the WASM-DSL language into Maude rewrite theories, ensuring automatic conformance with the WebAssembly specification.

BML Lab: Relation prediction of Disease and Transcriptome using LRP method — 2020 - 2021

- Participated in the Bioinformatics and Machine Learning Lab as an undergraduate researcher and founding member
- Developed a disease-transcriptome prediction model using PyTorch, integrating Layer-wise Relevance Propagation (LRP) to interpret model decisions. Optimized computational efficiency for experimental data processing

MimicStudio: Web service that automatically generates 3D product models — 2022

- Led the team and implemented the neural-rendering server with Instant-NGP
- Implemented an upscaling algorithm to improve the resolution of 3D videos

Cirkadian: Healthcare devices based on circadian rhythm — 2022.03 - 2022.08

- Developed a cross-platform mobile application using Flutter to interface with circadian illuminometers (light sensors) and control circadian lamps remotely
- Improved application stability and user experience (UI/UX) by resolving bugs and implementing iterative design improvements

COSMIC: AI-Driven Web Projects Club — 2020 - 2022

- Founded and led the club; built web projects including an AI-based security-camera service and an undergraduate course-recommendation platform

- Designed and supervised several instructional toy projects to teach web frameworks such as React, FastAPI and DJango

TEACHING

CSED415: Computer Security (TA)	2025 Spring
5110160: Machine Learning (TA)	2021 Fall
0622014: Applied Computer Programming (TA)	2020 Fall, 2021 Fall
0914002: Fundamental Computer Programming (TA)	2021 Spring, 2022 Spring

AWARDS & FUNDING

Master's Research Fellowship — National Research Foundation of Korea	2024–2025
Lotte Scholarship	2020–2023
Daeshin Freight Scholarship	2020–2023
Software Maestro Program Grant, 13th Cohort — Ministry of Science and ICT	2022
Excellence Scholarship — Software-Centered University Program, CBNU	2020–2022